

Since $r > 0$ the RHS of the above goes to $-\infty$ as $r \rightarrow \infty$.
Thus for sufficiently large $r > 0$.

$$v(\alpha, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0) \quad (\alpha \in \partial B_r(\mathcal{U}), t > 0)$$

Therefore

$$v(\mathcal{U}, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\mathcal{U}, 0) \quad (\mathcal{U} \in \partial B_r(\mathcal{U}), t > 0)$$

Therefore

$$v(\mathcal{U}, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0) \\ \therefore v(\mathcal{U}, t) \leq \sup_{\alpha \in \mathbb{R}^n} v(\mathcal{U}, t) \quad (\alpha, t) \in P_B(B_r(\mathcal{U}) \times [0, T - \varepsilon])$$

Finally taking $\mu \downarrow 0$

$$u(\mathcal{U}, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

$\therefore$ If $4aT < 1$ then

$$\sup_{(\alpha, t) \in \mathbb{R}^n \times [a, T]} u(\alpha, t) = \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

β) Let us consider general $T > 0$.

First take $T_1 = 1/8a$.

$$\Rightarrow \sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T_1]} u(\alpha, t) = \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

Next take $T_2 = 1/4a$

$$\sup_{(\alpha, t) \in \mathbb{R}^n \times [T_1, T_2]} u(\alpha, t) = \sup_{\alpha \in \mathbb{R}^n} u(\alpha, T_1) \\ \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

Chapter 15 1-dim wave equation and d'Alembert

$$\therefore \sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T_2]} u(\alpha, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

Repeating the above argument. we have

$$\sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T]} u(\alpha, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

Chapter 15: 1-dim wave eq and d'Alembert's formula.

$$(15.1) \begin{cases} \frac{\partial^2 u}{\partial t^2}(\alpha, t) - \frac{\partial^2 u}{\partial x^2}(\alpha, t) = 0, & \alpha \in \mathbb{R}, t \in \mathbb{R} \\ u(\alpha, 0) = 0, & \alpha \in \mathbb{R} \\ \frac{\partial u}{\partial t}(\alpha, 0) = h(\alpha), & \alpha \in \mathbb{R} \end{cases}$$

$g = g(\alpha) : \mathbb{R} \rightarrow \mathbb{R}$; given

$h = h(\alpha) : \mathbb{R} \rightarrow \mathbb{R}$; given

$u = u(\alpha, t) : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$; given unknown.

Problem: Derive representation formula for (15.1).

Observe that

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} \right) u \\ = \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x} \right) u$$

Thus, define.

$$v(\alpha, t) := \frac{\partial}{\partial t} u(\alpha, t) - \frac{\partial}{\partial x} u(\alpha, t) \quad (15.3)$$

Then from (15.1)

$$\frac{\partial^2 u}{\partial t^2}(x, t) - \frac{\partial^2 u}{\partial x^2}(x, t) = 0$$

$$\Leftrightarrow \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x} \right) v(x, t) = 0 \quad (15.4).$$

Now we look at (15.4) instead of (15.1)

< transport equation >

For $\vec{b} \in \mathbb{R}^n$, $f = f(x, t) : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$

the equation

$$(15.5) \quad \frac{\partial \phi}{\partial t}(x, t) + \vec{b} \cdot \nabla \phi(x, t) = f(x, t), \quad x \in \mathbb{R}^n, t \in \mathbb{R}$$

is called transport equation.

For fixed $x \in \mathbb{R}^n$ and $t \in \mathbb{R}$ put

$$\psi(s) = \phi(x + s\vec{b}, t + s), \quad s \in \mathbb{R}$$

Then

$$\begin{aligned} \frac{\partial}{\partial s} \psi(s) &= \frac{\partial \phi}{\partial t}(x + s\vec{b}, t + s) + \nabla \phi(x + s\vec{b}, t + s) \cdot \vec{b} \\ &= f(x + s\vec{b}, t + s) \quad (15.5). \end{aligned}$$

Integrate over $0 \leq s \leq \tau$.

we have

$$\int_0^\tau \frac{\partial}{\partial s} \psi(s) ds = \int_0^\tau f(x + s\vec{b}, t + s) ds$$

$$\parallel \psi(\tau) - \psi(0)$$

$$\parallel$$

$$\psi(\tau) - \phi(x, t).$$

Letting $\tau = -t$, we obtain

$$\begin{aligned} \phi(x - t\vec{b}, 0) - \phi(x, t) &= \int_0^{-t} \cancel{f(x + s\vec{b}, t + s)} f(x + s\vec{b}, t + s) ds \\ &= - \int_{-t}^0 f(x + s\vec{b}, t + s) ds \end{aligned}$$

hence.

$$\phi(x, t) = \phi(x - t\vec{b}, 0) + \int_{-t}^0 f(x + s\vec{b}, t + s) ds \quad (15.6).$$

(15.6) is the representation formula for transport equation (15.5)

In particular if $f \equiv 0$.

$$(15.7) \quad \phi(x, t) = \phi(x - t\vec{b}, 0) = a(x - t\vec{b}),$$

where $a(y) = \phi(y, 0)$.

< d'Alembert's formula >

Let u be a sol of (15.5). and let v defined by (15.3)

Recall.

$$(15.3) \quad v(x, t) = \frac{\partial u}{\partial t}(x, t) - \frac{\partial u}{\partial x}(x, t),$$

$$(15.4) \quad \frac{\partial v}{\partial t}(x, t) + \frac{\partial v}{\partial x}(x, t) = 0.$$

Since (15.4) is a transport equation with $n=1$, $\vec{b}=1$, $f \equiv 0$.

From (15.7) we have.

$$v(x, t) = a(x - t)$$

where $a(y) = v(y, 0)$.

Since (15.3) is a transport equation with $n=1$, $\vec{b}=-1$, $f=v$ we have.

$$\begin{aligned} u(x, t) &= u(x + t, 0) + \int_{-t}^0 v(x - s, t - s) ds \\ &= u(x + t, 0) + \int_{-t}^0 a((x - s) - (t + s)) ds \\ &= u(x + t, 0) + \int_{-t}^0 a(x - t - 2s) ds \end{aligned}$$

Taking the change of variable.

$$\tau = x - t - 2s$$

$$\int_{-t}^0 a(x-t-2s) ds = \frac{1}{2} \int_{x-t}^{x+t} a(\tau) d\tau$$

Thus

$$u(x, t) = g(x+t) + \frac{1}{2} \int_{x-t}^{x+t} a(\tau) d\tau.$$

Since

$$a(y) = v(y, 0)$$

$$= \frac{\partial u}{\partial t}(y, 0) - \frac{\partial u}{\partial x}(y, 0)$$

$$= h(y) - \frac{\partial g}{\partial x}(y)$$

we have

$$\int_{x-t}^{x+t} a(\tau) d\tau = \int_{x-t}^{x+t} \left(h(\tau) - \frac{\partial g}{\partial x}(\tau) \right) d\tau$$

$$\begin{aligned} &= \int_{x-t}^{x+t} h(\tau) d\tau - \left[g(\tau) \right]_{x-t}^{x+t} \\ &= \int_{x-t}^{x+t} h(\tau) d\tau - (g(x+t) - g(x-t)) \end{aligned}$$

Therefore

$$u(x, t) = \frac{1}{2} (g(x+t) - g(x-t)) + \frac{1}{2} \int_{x-t}^{x+t} h(\tau) d\tau \quad (15.11)$$

(15.11) is called d'Alembert formula.

Thm 15.1: For $g \in C^2(\mathbb{R})$, $h \in C^1(\mathbb{R})$, let u define (15.11).
Then $u \in C^2(\mathbb{R} \times \mathbb{R})$ is a solution of (15.1).

< Difference between heat eq and wave eq >

• smoothness / regularity of sol.

Thm 12.2 (heat eq.) $u \in C^\infty(\mathbb{R} \times (0, \infty))$.

Thm 15.1 (wave eq.) $u \in C^2(\mathbb{R} \times \mathbb{R})$

Heat eq $\rightarrow$ solution is smooth immediately.

Wave eq $\rightarrow$ Regularity of sol = Regularity of the initial data.

• solve to past time.

Thm 12.2 (Heat eq.) $u \in C^\infty(\mathbb{R} \times (0, \infty))$.

Thm 15.1 (Wave eq.) $u \in C^2(\mathbb{R} \times \mathbb{R})$ time

Heat eq $\rightarrow$ cannot solve to past time in general.

Wave eq $\rightarrow$ can solve to past time.

• Finite / Infinite propagation.

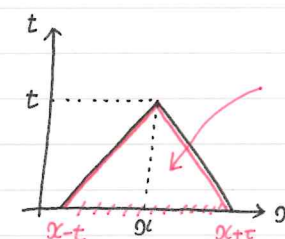
Thm 12.2 (Heat eq.)

$$u(x, t) = \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}} e^{-\frac{|x-y|^2}{4t}} g(y) dy, \quad x \in \mathbb{R}, t > 0$$

Infinite propagation (無限伝播性).

Thm 15.1 (Wave eq.)

$$u(x, t) = \frac{1}{2} (g(x+t) - g(x-t)) + \frac{1}{2} \int_{x-t}^{x+t} h(\tau) d\tau$$



to determine $u(x, t)$
we need the data on this region

finite propagation (有限伝播性).